Arush Nandakumar Menon

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EXPERIENCE

CGI Jun 2026 – Jul 2026
Software Intern Bengaluru, India

- Engineered an insurance pricing engine achieving an R^2 of 0.97 on 2,000+ cases by deploying a scikit-learn Gradient Boosting model via a FastAPI REST API.
- Built a deterministic rules engine in Next.js and TypeScript evaluating 28+ medical conditions and 11 riders, generating instant 0–100 risk profiles and PDF quotes.

Startrit Infratech Private Limited Dec 2025 – Mar 2026
Software Intern Remote, India

- Spearheaded a full-stack platform spanning React/TypeScript, Node.js/Express, PostgreSQL/Supabase, and real-time Socket.IO infrastructure.
- Integrated Python/FastAPI AI services for semantic search, automated scoring, and document processing workflows.

TECHNICAL SKILLS

Programming Languages	Java, C/C++, Python, JavaScript, TypeScript, SQL
Frameworks & Libraries	React, Next.js, Node.js, Express, Flask, FastAPI, Tailwind CSS, Zod, Socket.IO
AI / ML	RAG, OCR, Ollama, scikit-learn, PyTorch, TensorFlow, YOLOv8, OpenCV
Databases & Cloud	PostgreSQL, Supabase, Firebase, Vercel, MongoDB
Tools / DevOps	Git, GitHub, Docker, Postman, GitHub Actions, CI/CD

PROJECTS

Where's My Offer: Placement Tracker | *Next.js, PostgreSQL, Gmail API* | GitHub | Live Aug 2026 – Ongoing

- Developed a real-time placement tracker, syncing 1,000+ emails in <1s with 98% accuracy, via dual-inbox OAuth 2.0, Pub/Sub webhooks, and a 30+ rule classifier.
- Accelerated shortlist verification, scanning 4,000+ row spreadsheets in <200ms, by developing an in-memory XLSX parser with ZIP validation and regex ID matching.
- Eliminated sync race conditions, ensuring 100% consistency across 5 stages, via atomic PostgreSQL lease locks.

StudentHub: AI Study Platform | *Next.js 15, Supabase, Gemini AI* | GitHub | Live Jan 2026 – May 2026

- Architected a 15+ module system across 30 REST APIs, isolating multi-tenant data on a 32-table Supabase schema with RLS and Zod.
- Constructed a client-side PDF/OCR pipeline, extracting data at 98% precision with no server-side expense, via pdf.js and 3px row clustering.
- Automated AI study features, cutting repeat API calls via SHA-256 caching and jsonrepair.

LocalMind: Hybrid AI Assistant | *Next.js, Python, Ollama, Gemini* | GitHub | Live Jul 2025 – Jan 2026

- Reduced cloud LLM costs by 85% with zero downtime by architecting a hybrid Flask gateway routing 7 local Ollama models with a 4s Gemini 2.5 failover.
- Lowered agent prompt overhead by 40% by designing an observe-decide-act loop that prunes a 10-tool registry to top 6 tools, paired with a wake-word voice UI.
- Trimmed context token usage by 65% via a 16-turn sliding summarizer and pgvector RAG, backed by 17 Jest tests.

EDUCATION

Vellore Institute of Technology Sep 2023 – Jun 2027
B.Tech in Computer Science and Engineering | CGPA: 8.01/10 Sehore, India
Relevant Coursework: DSA, Operating Systems, Computer Networks, AI/ML Fundamentals

EXTRACURRICULAR & LEADERSHIP

The Electric Vehicle Club, VIT Jan 2024 – May 2025
Technical Lead Sehore, India

- Led technical operations for 3 events serving 500+ students, coordinating logistics, setup, and Rs.5K+ prize pools.
- Created club website with JavaScript and GSAP, enabling event discovery and self-service certificates and appointments.

CERTIFICATIONS & LANGUAGES

- **Certifications:** NPTEL: Intro to ML, NPTEL: Marketing Analytics (Elite, 91%), Google/Coursera: Computer Networking, Apna College: Delta Web Development, AWS Cloud Practitioner Essentials
- **Languages:** English (Fluent), Malayalam (Native), Hindi (Fluent), Tamil (Intermediate), Kannada (Basic)